



Teamcenter Unified Architecture

Product Configurator

(User Manual)

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1) About this Guide

This guide provides overview of TCUA - Product Configurator application and features to explain how to use the basic functionalities of Product Configurator. It is intended for end users of TCUA. It will allow a person who is new to this application to learn the steps to be followed while using basic features like create configurator context, SVRs etc.

2) Scope of the Document

The purpose of the document is to provide an overview of the recommended procedures to follow while using TCUA Product Configurator.

3) Introduction

3.1 Overview of PLM and Teamcenter Unified Architecture

What is PLM?

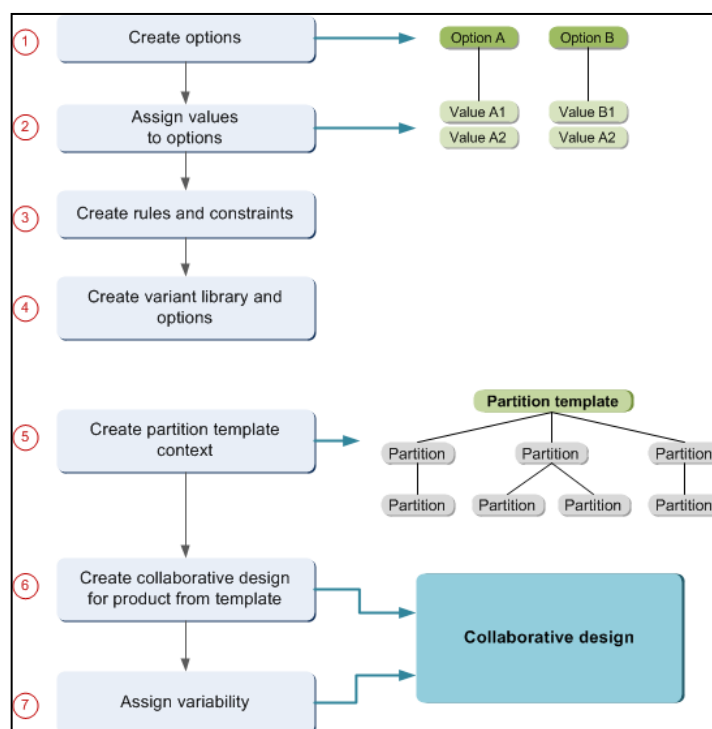
PLM is a strategic business approach that applies a consistent set of business solutions that support the collaborative creation, management, dissemination, and use of product definition information. It is not just a technology, but is an approach in which processes are as important.

Teamcenter Unified Architecture (TCUA) is Siemens Teamcenter tool that is developed for Product Lifecycle Management. The tool is designed to manage the entire lifecycle of a product from its conception, through design and manufacture, to service and disposal.

3.2 Introduction to TCUA- Product Configurator

Product Configurator is the part of Siemens Teamcenter Unified Architecture tool.

Product Configurator helps to manage various product models. It allows managing variability and helps to ensure consistency while authoring and qualifying the product data. With Product Configurator, we can define and manage variability for a product right from the beginning i.e. requirements gathering till the end where an order is delivered to a customer.

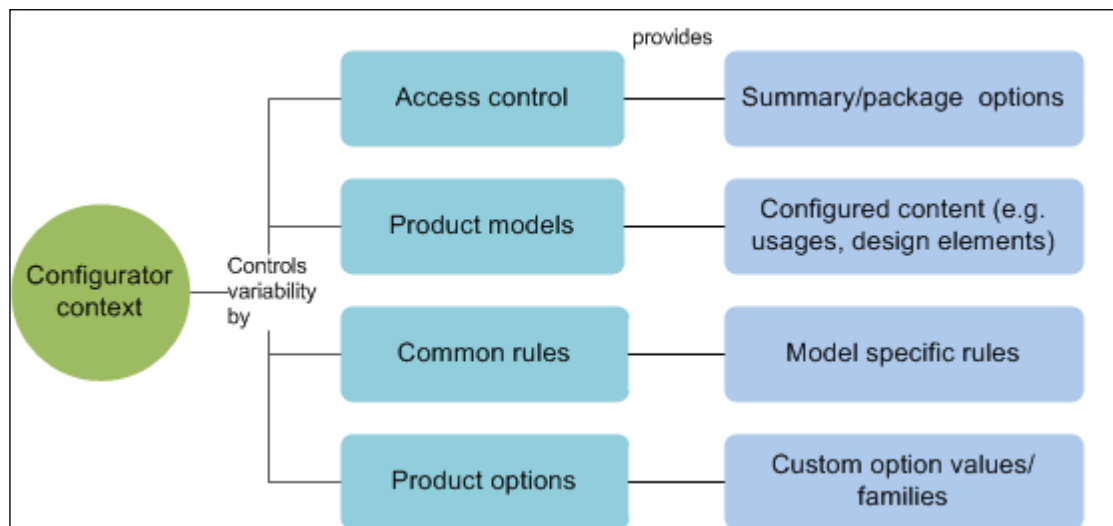


4) Important Terminologies in TCUA Product Configurator

4.1 Configurator Context

A configurator context represents a family of products, such as a product line, that may share the same set of option family groups, option families, and option values.

For example, cars and vans may be two different product lines that can be represented by two separate configurator contexts.



4.2 Option Family

Option Family is a method of grouping option values with a similar purpose. For example, you can create an option family called color with option values of red, green, and yellow. An option family may be allocated to configurator contexts.

4.3 Option Value

An option value represents a possible value in an option family. Option values are workspace objects with their own names, description, and owner. An option value belongs to one, and only one, option family. Option values may be freeform or a member of an enumerated list.

4.5 Configurator Rules

Configurator rules or constraints helps to specify which options and their combinations are allowed for a given product. There are 4 types of rules - Default, Inclusion, Exclusion and Applicability.

4.4 Saved Variant Rule

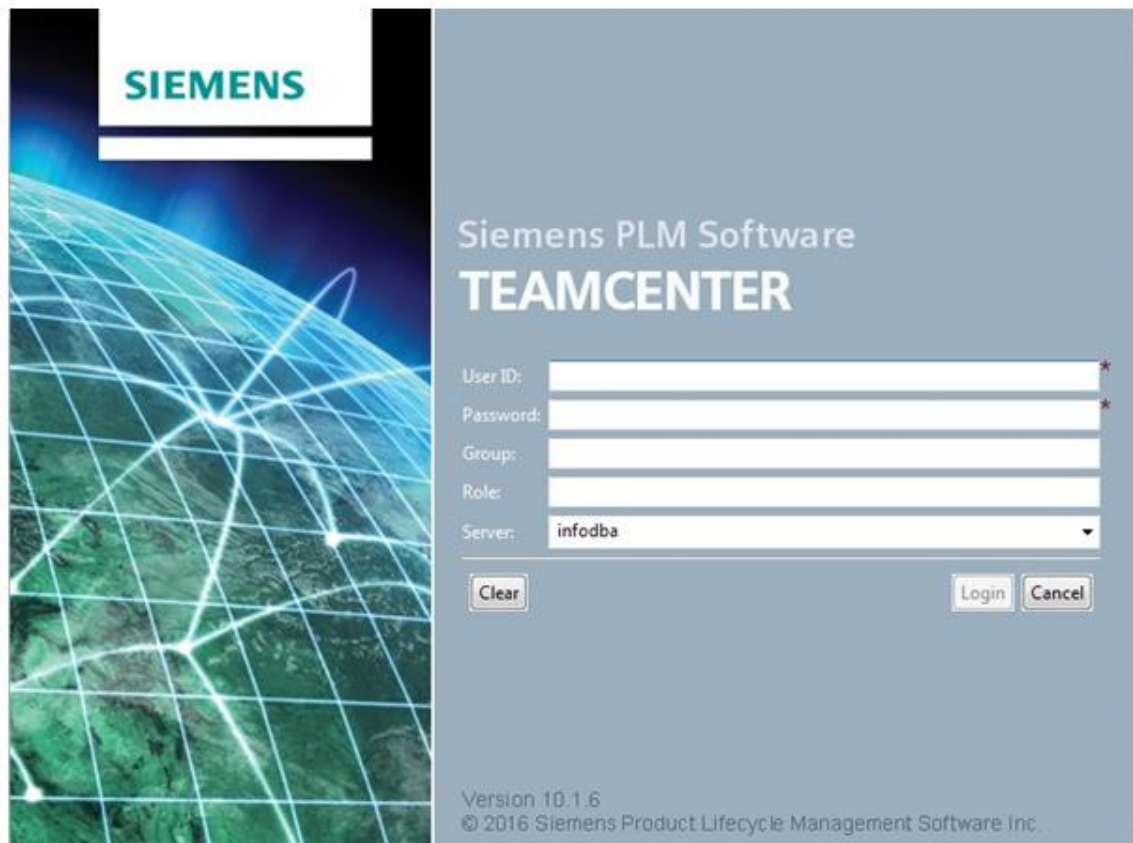
Save Variant Rule is an expression that enumerates a selection of option values, which may represent a complete or partial configuration. A variant rule must be validated as complete and consistent with no violations to produce a buildable physical variant.

5) Using Product Configurator

5.1 Start Teamcenter

Step 1: Go to installation location of Teamcenter (C:\Siemens\infodba\rac), and click on portal.bat

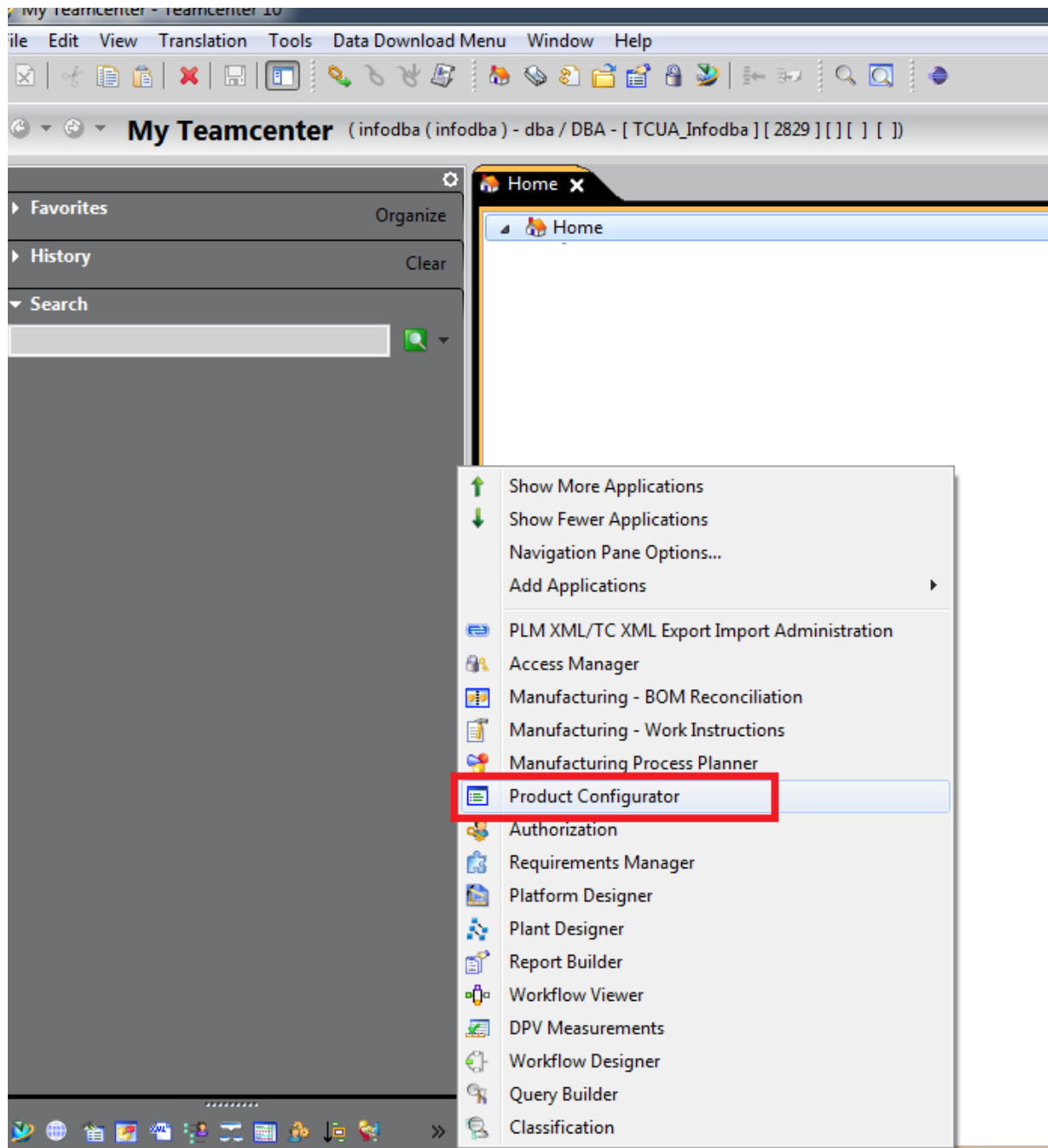
Step 2: Following window will pop –up .Enter Username and Password and login to TCUA client



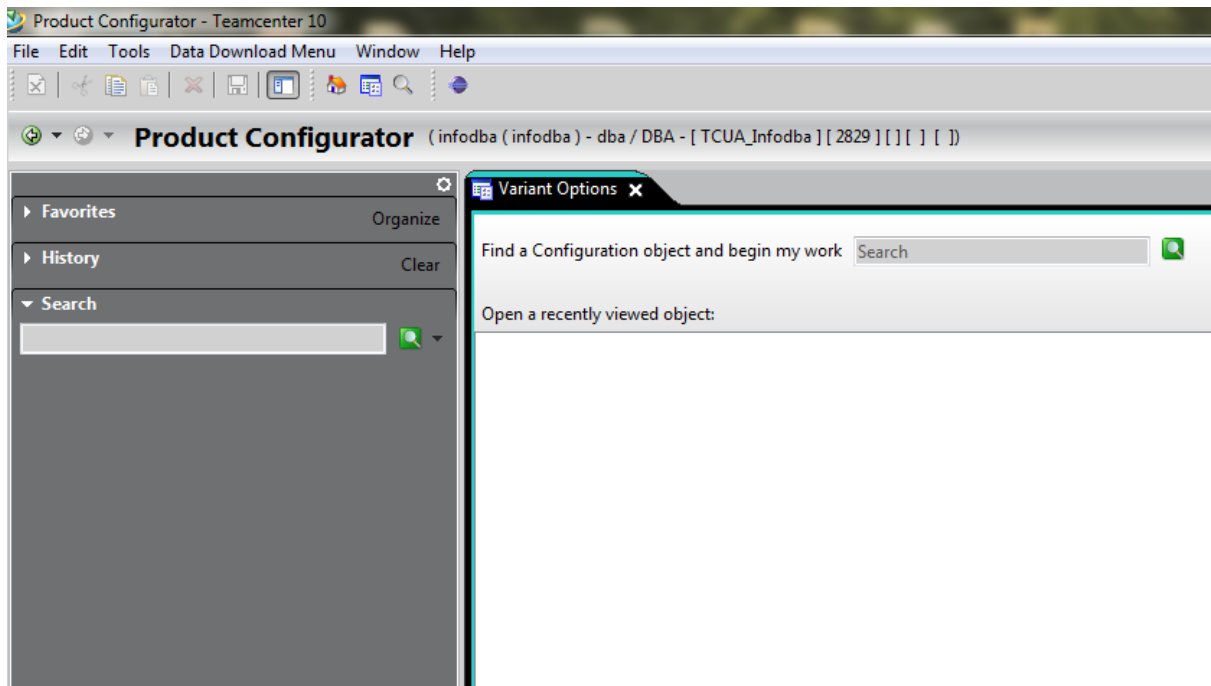
Step 3: Enter userid and password and press login. TCUA Getting Started Window will get launched.

5.2 Launch Product Configurator

Step 1: Select Product Configurator from the left side panel as shown below.



Product Configurator perspective will get opened as shown below



5.3 Create Configurator Context

Step 1: Go to menu File → New → Configurator Context in Product Configurator perspective. Window as shown below will appear.

Step 2: Click assign to auto assign id and revision.

Step 3: Enter Name and description.

Step 4: Click Finish.

New Business Object

Object Create Information
Define business object create information

Configurator Context General

▼ **Item Information (required)**

ID:

Revision:

Name*:

Description:

► **Additional Item Information**

► **Item Revision Information**

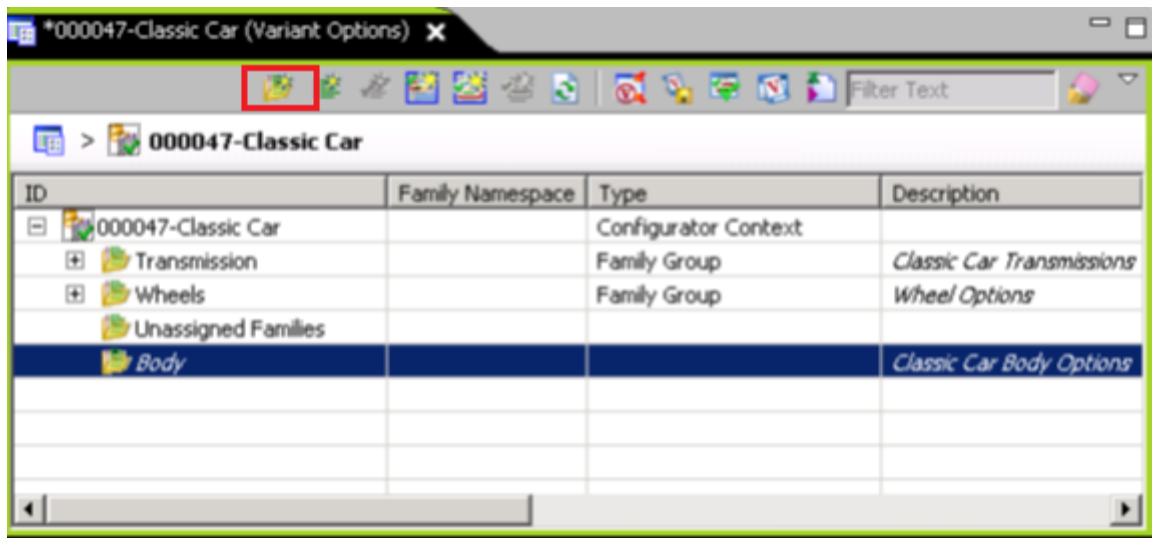
☐ Open On Create

< Back Next > **Finish** Cancel

5.4 Create Group, Options Families and Values

- To create Group -
 - Step 1: Click the button highlighted in Red below to create Group.
 - Step 2: A row under Configurator context will become editable.
 - Step 3: Enter Name and description.

Step 4: Click Save button on main toolbar to save changes.



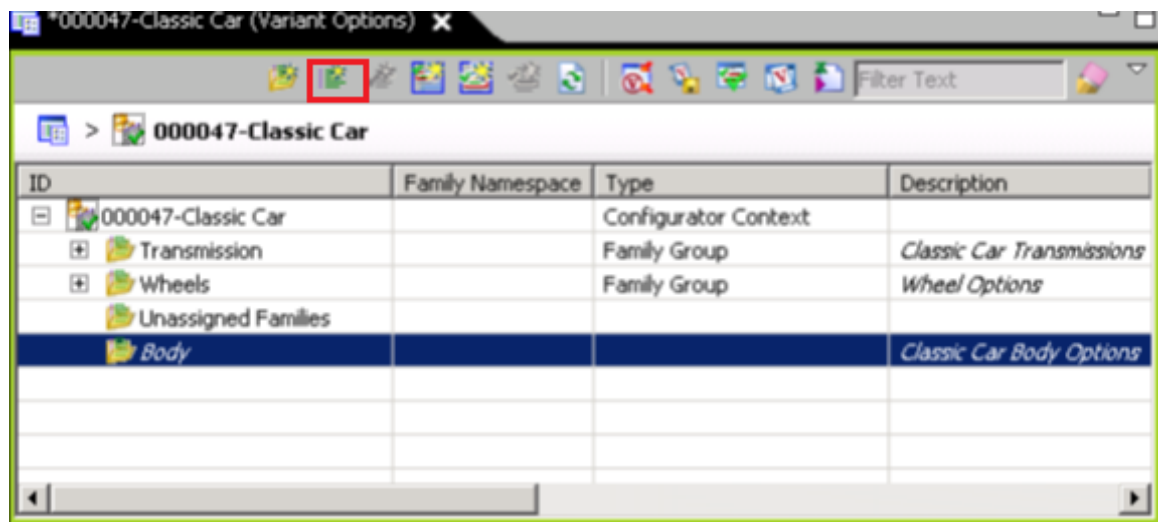
- **To create Option Family -**

Step 1: Select Group.

Step 2: Click the button highlighted in Red below to create option family. A row under Group will become editable.

Step 3: Enter Name and description.

Step 4: Click Save button on main toolbar to save changes.



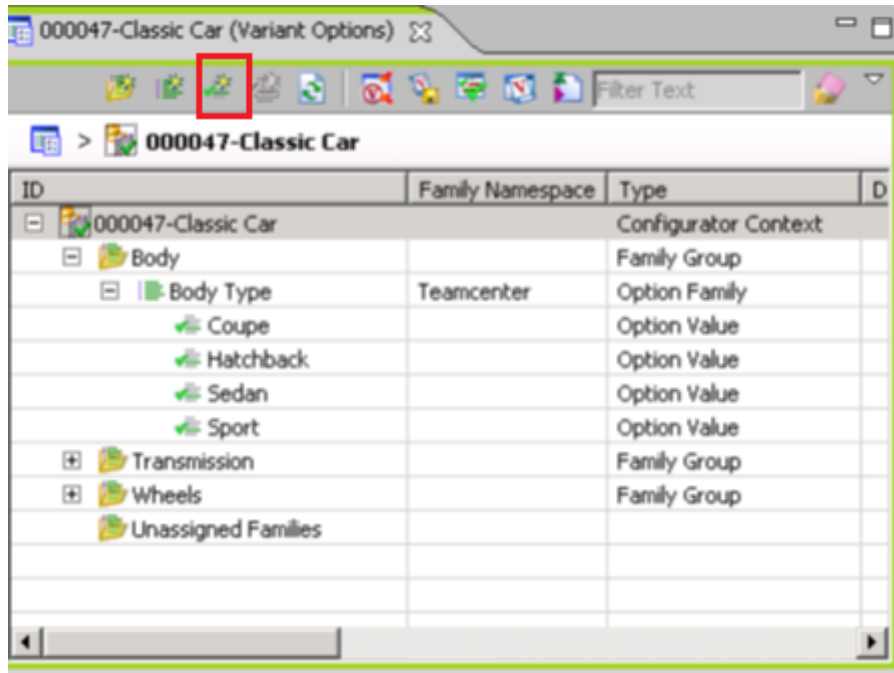
- **To create Option Values -**

Step 1: Select Option Family.

Step 2: Click the button highlighted in Red below to create option family. A row under Group will become editable.

Step 3: Enter Name and description.

Step 4: Click Save button on main toolbar to save changes.



5.5 Create Rules

Configurator rules define errors, warnings, and informative messages that Teamcenter generates if an invalid selection of variant option values is made.

Configurator rules are of three types:

- Default: Default rules specify default option values to use during configuration.
- Inclusive: Inclusion rules are met when the condition is satisfied.
- Exclusive: Exclusion rules are met when the condition is not satisfied.

To create the configurator rules -

Step 1: Open the configurator context in the Variant Options view, and then click Open

Configurator Rules view  to open the configurator rules editor.

Step 2: Click in the ID field of the empty row and type the ID of the rule, or leave it blank so that the next available ID is automatically assigned.

Step 3: Click the Type field in the same row. Select the Type (Default Rule, Exclusion Rule, or Inclusion Rule).

Step 4: Click the Severity field in the same row.

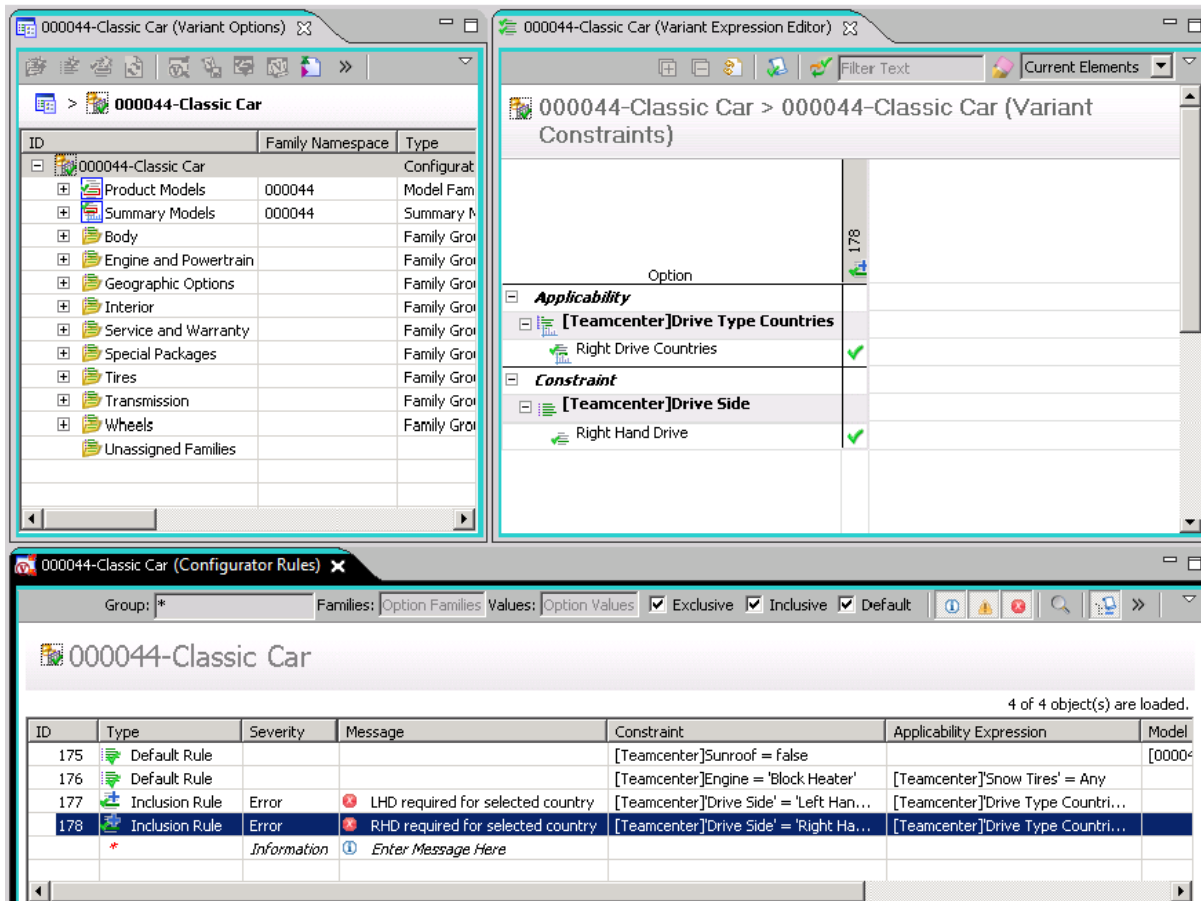
For inclusion and exclusion rules only:

- Select a severity (Error, Information or Warning).
- Click the Message field of the same row and type the text of the message.

Step 5: Click Save  on the main toolbar to save this newly created rule.

Step 6: Click  to display Variant Expression Editor.

Step 7: Define an expression for the configurator rule.



Step 8: Repeat steps 2 to 7, to create additional variant constraints, as required.

Step 9: Click  Save on the main toolbar to save changes.

If Teamcenter detects any violations of configurator rules, the violation error message appears. The configurator rule is displayed with a warning symbol and you can view a tooltip with a violation message if you hover the cursor over the failed rule.

5.6 Create Saved Variant Rules

Saved variant rules define the option values, which users have selected and saved for a given configuration.

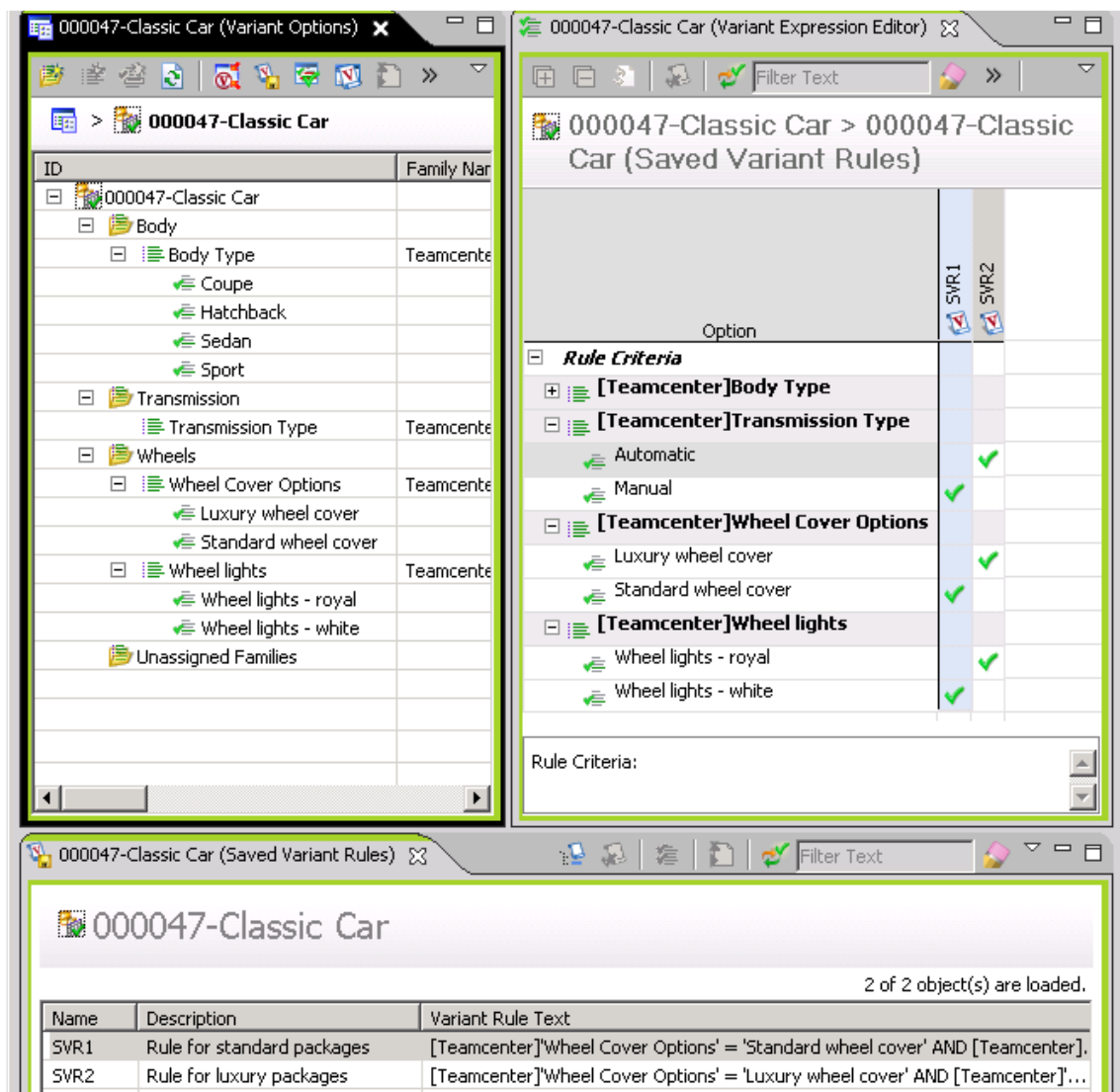
Step 1: Open the configurator context in the Product Configurator.

Step 2: Click icon  in variant option view to add a new saved variant rule (SVR).

Step 3: Type a name and description of the rule.

Step 4: Click . Teamcenter displays the Variant Expressions Editor view.

Step 5: Construct a variant expression for the selected rule in Teamcenter, as shown in the following image.



Step 6: Follow the step 4 and 5 for other saved variant rules.

Step 7: Click Save  on the main toolbar to save all additions, deletions, and edits to the database.

6) Applying Configurator Context to get configurable BOM

6.1 Attach Configurator Context To Product

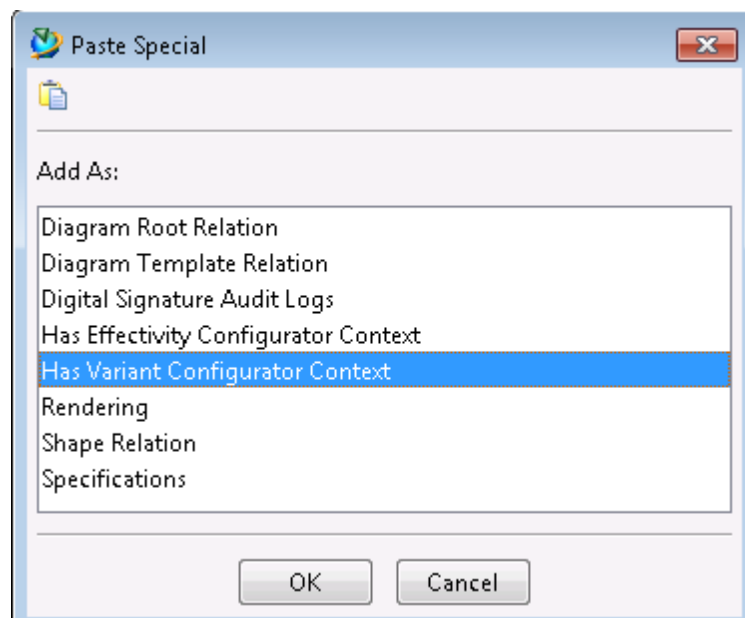
Step 1: Create Configurator context in Product Configurator. Refer section - 5 for detail steps of how to create configurator context.

Step 2: Select the context item. Goto Edit-> Copy.

Step 2: Open my Teamcenter. Refer Annexure -1 for details of My Teamcenter.

Step 3: In My Teamcenter, Query the Product Model or Root Node of the BOM to be configured.

Step 4: Go to option Edit→Paste Special. Then, choose Has Variant Configurator Context.



6.2 Define Variant Expressions

Step 1: Complete the steps mentioned in 6.1.

Step 2: Open the BOM (which is to be configured) in Structure Manager. Refer Annexure – 2 for details of Structure Manager.

Step 3: Select child item and click button  from toolbar. Variant expression editor window will appear.

Step 4: Add variant conditions for all child items as required.

000337/A;1-Product_CAR_V1 (Variant Expression Editor) X

000337/A;1-Product_CAR_V1 > 000337/A;1-Product_CAR_V1 (Saved Variant Rules)

	Luxury	Sports	Base
Model			
Base			✓
Sports		✓	
Luxury	✓		
Style			
Coupe			✓
Sedan	✓	✓	
Engine			
E1, Base Engine			✓
E2, V8	✓		
E3, V8 Turbo		✓	
Transmission			
T1, Manual		✓	✓
T2, Auto	✓		

Model = 'Luxury' AND Style = 'Sedan' AND Engine = 'E2, V8' AND Transmission = 'T2, Auto'

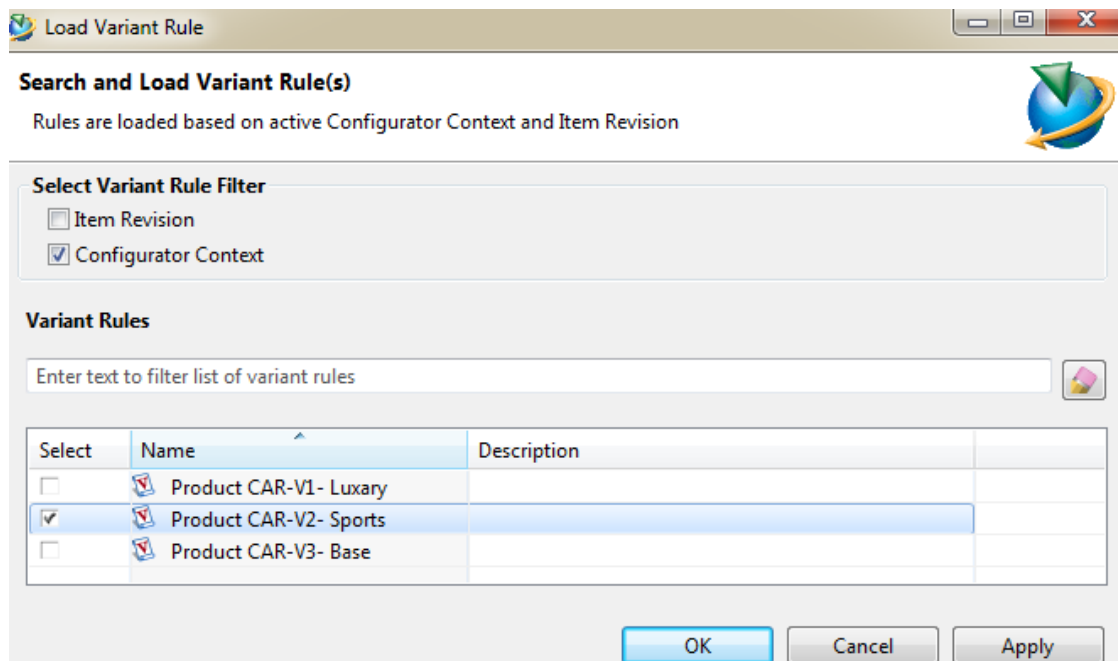
6.3 Apply Saved Variant Rule (SVR)

Step 1: Complete the steps mentioned in 6.2.

Step 2: In Structure Manager, in menu View, ensure that option 'Enable Variant Configure To Load' is highlighted. If not, click on the option once.

Step 3: Open the BOM (which is to be configured) in Structure Manager. Refer Annexure – 2 for details of Structure Manager. If BOM is already open in Structure Manager, click the icon  from main toolbar.

Step3: List of Saved Variant Rule will appear. Select SVR, which is to be applied. Structure Manager will show only configured variants applicable to that SVR.



Step4: To view unconfigured child items, select option 'Show Unconfigured Variants' from menu View.

6.4 View JT for Configured BOM

To view JT of individual component, use Graphics tab of Structure Manager Data Panel. Refer Annexure II for details of Structure Manager.

To view JT for an assembly, select item and click  button from toolbar. JT will be opened in Teamcenter Visualization application.

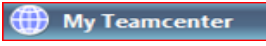
7) Annexures

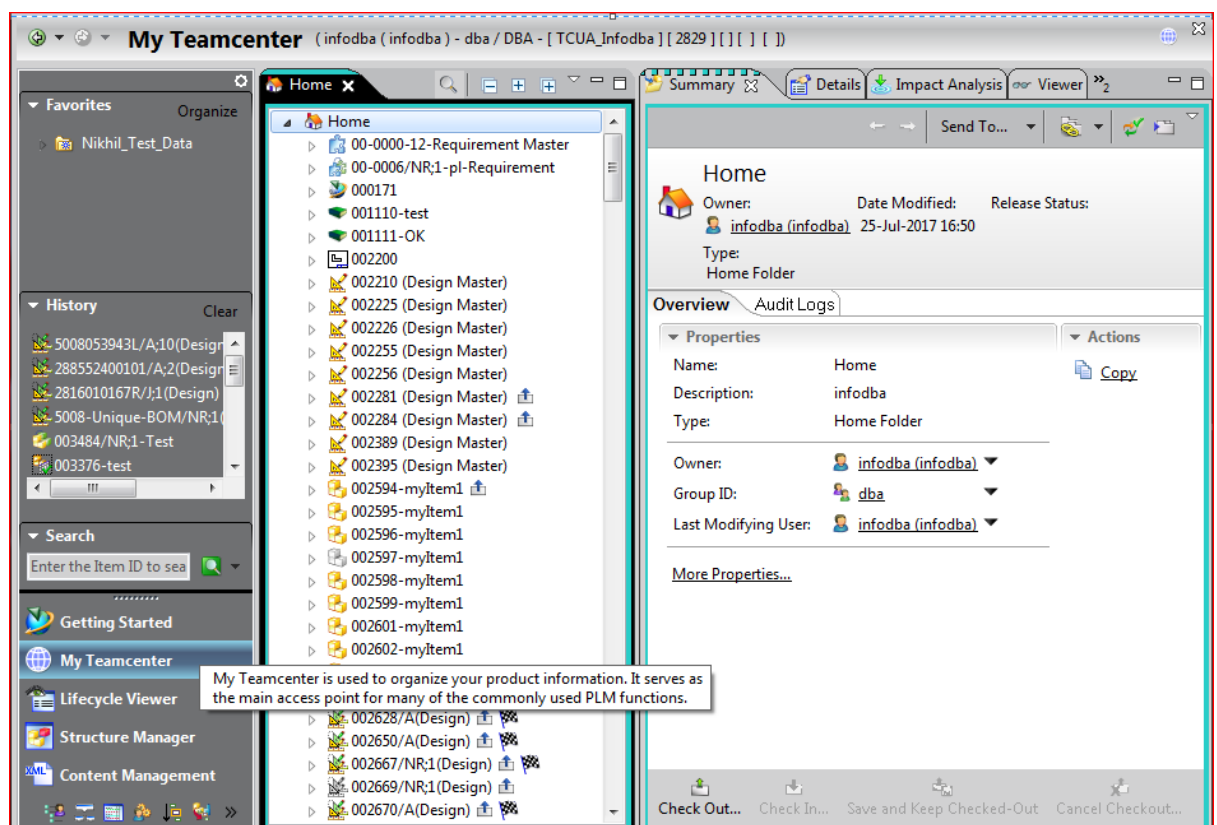
I. My Teamcenter

It provides Teamcenter Component, Summary, Details, Viewer, and Impact Analysis views that you use to perform many basic tasks such as viewing and modifying your data, initiating and participating in a workflow process, and checking your data into and out of the Teamcenter database.

- Open MyTeamcenter

Step 1: Launch Teamcenter.

Step 2: click My Teamcenter  in the navigation pane. My Teamcenter perspective will open as shown in below image.



- Query the Design revision

Step 1: Goto Search pane. Default search will be through Item ID. If not then please select Item ID by clicking .

Step 2: Enter the Item ID and click search .



II. Structure Manager

Structure Manager is Teamcenter application that enables creation of generic product structures that can be configured to show which product structures are in production, effective on a certain date, etc.

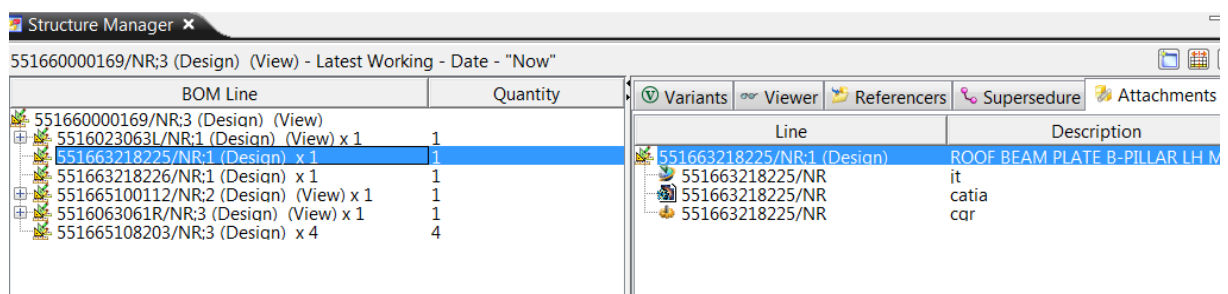
Structure Manager enables creation and modification of a product structure and its associated occurrence data, display of a product structure in a multilevel indented format, and viewing graphics tightly coupled to the structure for easy identification of a component by location in the structure or in the embedded viewer.

- Open Structure in Structure Manager

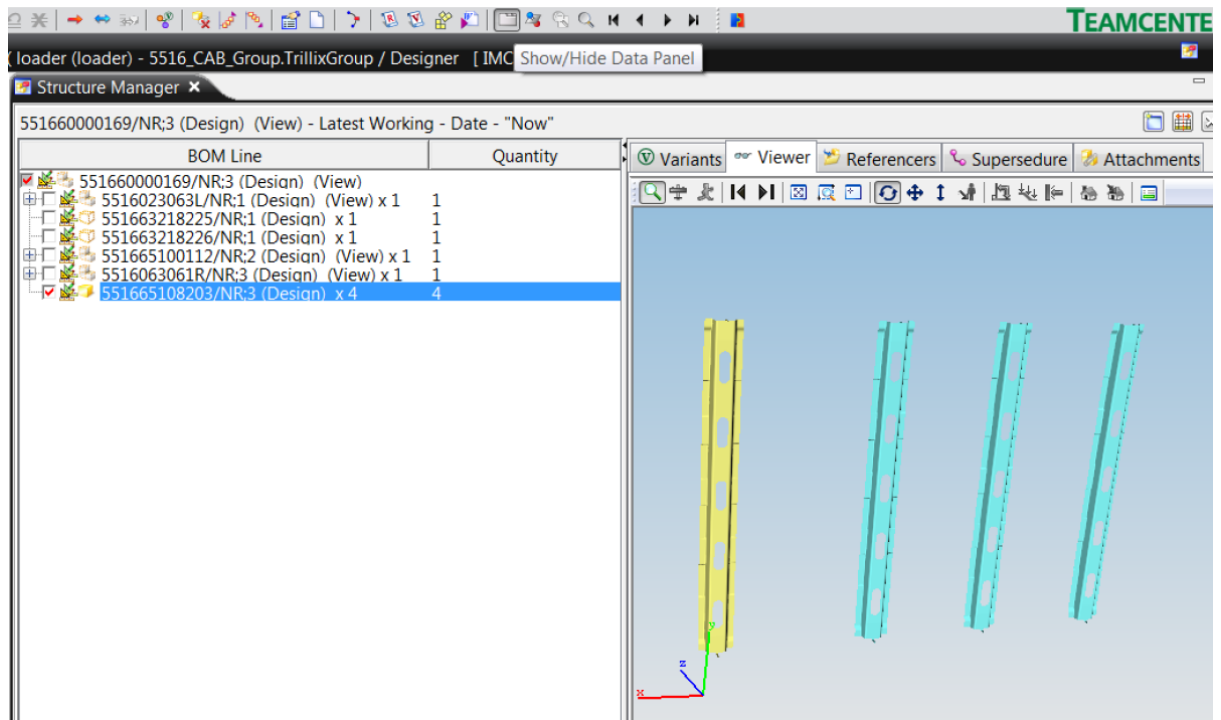
Step 1: Query the DesignRevision in MyTeamcenter.

Step 2: Select DesignRevision and right click. Click Send to -> Structure Manager.

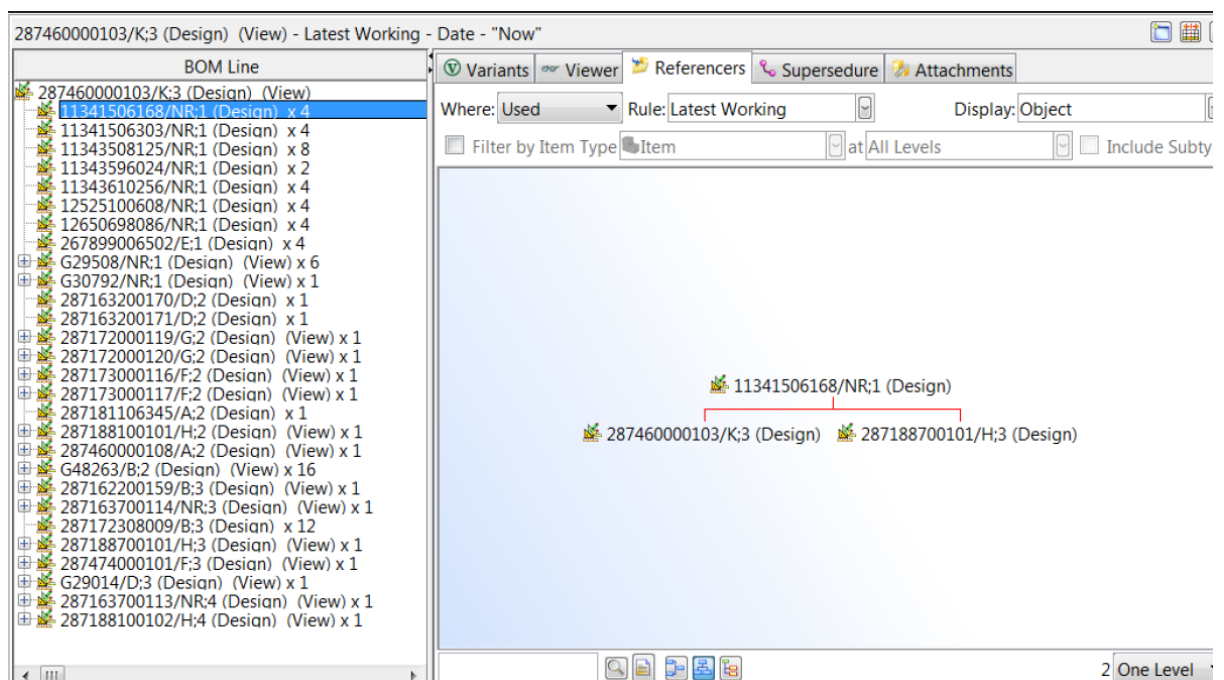
Step 3: Structure Manager will open the BOM for that DesignRevision. By default 1st level child item are shown. Use option Expand or Expand below to view all levels or to desired level respectively.



- Data Panel Features
- Viewer tab allows you to see JT files attached to a design



- References tab allows identifying the assemblies where the design is used.



- Attachments tab allows us to see all the datasets attached to Design.

